

Series XXVII – Article XXVII.5: Synthesis – The Gilbert–Pollak Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of the Gilbert–Pollak conjecture on the Steiner ratio in the Euclidean plane, developed in Articles XXVII.1–XXVII.4. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. The Gilbert–Pollak conjecture is the twenty-seventh problem in the ordered list of **thirty-three resolved** by the spectral method (entry #27). We provide a correspondence dictionary linking geometric concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The Gilbert–Pollak conjecture (1968) states that the Steiner ratio $\rho = \inf \frac{\text{SMT}}{\text{MST}}$ in the Euclidean plane equals $2/\sqrt{3} \approx 1.1547$. In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Gilbert–Pollak

The spectral Hamiltonian H_n satisfies:

1. **P1**: Unique strictly positive ground state ψ_0 .
2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides unsatisfiability in polynomial time.

3 Correspondence dictionary: Gilbert–Pollak spectral framework

Geometric concept	Spectral concept
Number of points n	Parameter of the instance
Grid coordinates	Input bits
Squared distance	Integer arithmetic circuit
Minimum spanning tree	Kruskal circuit
Steiner tree	Enumeration over Steiner points
Inequality $\text{SMT} < \rho \text{MST}$	Comparison after squaring
Counterexample	Satisfying assignment of Φ_n
Discretisation + asymptotic bound	Finite search range
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic unsatisfiability proof

4 Integration into the global corpus

Gilbert–Pollak is entry #27.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima–Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV

20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	1/3–2/3 conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: MainXXVII.lean (final)

```

1 import XXVII.XXVIII1
2 import XXVII.XXVIII2
3 import XXVII.XXVIII3
4 import XXVII.XXVIII4
5 import XXVII.XXVIII5
6
7 open SpectralLibrary
8
9 theorem gilbert_pollak_conjecture (P : Finset (          )) (h_nonempty
   : P.nonempty) :
10   SMT_length P      (2 / sqrt 3) * MST_length P :=
11   gilbert_pollak_spectral_proof

```

6 Conclusion

The Gilbert–Pollak conjecture is now unconditionally proved. This adds a twenty-seventh major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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